

JOEL SUSHANTH MARKAPUDI

DATA ENGINEER | DATA & MACHINE LEARNING ENGINEER

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PROFESSIONAL SUMMARY

Data & Machine Learning Engineer with four years of full-time experience building enterprise data platforms, FP&A analytics, and financial decision models. Delivered reporting, cross-system reconciliation, calibrated ML, and production BI across post-acquisition operations.

PROFESSIONAL EXPERIENCE

Data & Machine Learning Engineer | Innova Solutions | Hyderabad, India

May 2022 - Nov 2023

Financial Machine Learning | Python, XGBoost, scikit-learn, PyTorch, Optuna

- Built governed datasets for financial ML products, defined grain, point-in-time features, leakage controls, and traceable labels.
- Developed rate and margin guidance using XGBoost quantile regression, cutting bill-rate error from \$6.20/hr to \$3.65/hr with 81.2% coverage for an intended 80% prediction range.
- Built assignment attrition and redeployment risk using a discrete-time XGBoost hazard model, lifting two-month AUC from a 0.5554 tenure baseline to 0.7067. Applied isotonic calibration to align risk scores with observed termination rates for reliable worklists and value estimates.
- Built a collections late-payment classifier and a cash-timing hazard model, reconstructing payment windows from monthly AR snapshots. Reduced late-payment probability error by 32% and reached 0.7958 AUC for payment timing.
- Built revenue-assurance controls that separated systematic accounting exceptions from rare anomalies, using deterministic checks, Isolation Forest, and an autoencoder. Surfaced 45,685 lines carrying \$37.17M in revenue against near-zero recorded pay for finance review.
- Owned Power BI platform administration across two tenants, gateway upgrades, and server migrations. Partnered with finance leaders, conducted 35+ hiring interviews, mentored junior engineers, and earned the Innova Idol Award.

Data Engineer | Innova Solutions | Hyderabad, India

Dec 2019 - Nov 2023

FP&A & Operational Analytics | Python, Talend, PL/SQL, Power BI, DAX, Oracle ADW

- Led BI delivery across 11+ finance and operations programs, including QBR, Producer Commission, Daily Sales & Outstanding, and SG&A, translating stakeholder requirements into reporting through Python, Talend, and PL/SQL pipelines.
- Shipped 25+ dashboards and Power BI semantic models with 40 tables, 187 relationships, and 354 measures. Automated margin and audit workflows, eliminating 70+ hours/month of manual work. Reduced deployment time by over 50% through CI/CD improvements.
- Built DAX YTD actual/projection forecasts and a ranking eligibility engine using GP projections and placements.
- Designed revenue-crediting and amortization rules that prevented double-counting across roles, while preserving individual scorecard visibility.
- Reconciled employee, producer, client, and assignment identities across six systems without shared keys using 6+ layered match patterns, collision review, residue tracking, and manual overrides.
- Delivered 300+ operational KPIs with forecasting, actuals-vs-baseline tracking, hierarchical rollups, drill-down analysis, and financial RCA.

EDUCATION

Northeastern University

M.S., Artificial Intelligence · GPA 3.9

Jan 2024 - Dec 2025

Boston, MA

Koneru Lakshmaiah Education Foundation

Bachelor's, Computer Science · GPA 9.0

Jun 2016 - Jun 2020

India

TECHNICAL SKILLS

Programming & ML: Python (PyTorch, pytorch-cuda, Polars, NumPy, OpenCV, XGBoost), SQL, PL/SQL, Java, DAX, Javascript, Typescript
Data Engineering & BI: Oracle ADW, SQL Server, DuckDB, Postgres, Talend, Power BI, Power Query, Tabular Editor, Airflow
AI & Applications: LangGraph, LangChain, MCP, FastAPI, Streamlit, MLX-LM, Ollama, vLLM, LlamaIndex, Optuna, MLflow
Cloud & Delivery: AWS Bedrock, S3, Lambda, ECS, Docker, CI/CD, Kubernetes, GitHub Actions, Pytest, Ruff, ELK

PROJECTS

FinSights: LLM RAG Financial Data Intelligence [\[GitHub\]](#)

Sep 2025 - Dec 2025

- Built an end-to-end financial RAG system over SEC 10-K filings (25 companies), using custom prepared KPI tables, S3 Vectors and DuckDB. Generated the vector corpus in three bins, with checkpointed fail-safe embedding patterns, at 1,850 vectors/min for \$2.21.
- Engineered ten retriever control algorithms like entity adaptation, metadata filtering, and variant generation into one modular pipeline.
- Deployed FastAPI/Streamlit orchestration on ECS Fargate (3 GiB, sized against a 1.22 GiB peak). Analyzed a Bedrock throttle spiral to boto3's hidden retry layer and built a fixed-budget rate limiter against an 8.1M-token/day quota.
- Validated retrieval with deterministic Hit@k and MRR tests, then evaluated answer quality with BERTScore, BLEURT-20, ROUGE-L, and manual gold-test reviews. A custom 30-question suite averaged 0.826 BERTScore F1.
- Profiled per-request component rebuild cost (1.16s), then cached it out entirely. Implemented progressive disclosure and live SSE streaming through a queue-bridge pattern, achieving 4.3 ms measured HTTP TTFB.

Local Multi-Agent LLM Debate System (LangGraph, MCP, MLX) [\[GitHub\]](#)

Sep 2025 - Dec 2025

- Architected a debate pipeline as a 13-node LangGraph state machine with typed state, conditional routing, SQLite checkpointing, fan-out for judging, and deterministic crash recovery. Benchmarked BM25, dense, and hybrid retrieval over 14,195 Touché-2020 passages.
- Exposed the grounding layer through two MCP servers with 9 tools and 7 resources.
- Served ~34 GB of quantized MLX weights on 24 GB memory, using a lock-protected, phase-aware residency manager that serialized model access and evicted weights between debate and judging.